

A n s w e r s

1. **c.** Section 8 holds administrative commands such as fsck and mkfs.
2. **a.** The !! command history expansion executes the previous command. Entering the Ctrl-P keystroke uses the Emacs key-binding bash to move up one line in the history; pressing Enter executes that command.
3. The man command displays manpages.
4. **b and c.** find and locate do not search the contents of files. /etc/passwd is not a script.
5. **a.** The variable must be set and exported. The semicolon separates the two commands.
6. **/etc/group.**
7. **c.** The g indicates that we're operating on the group privilege, and the +s indicates that we should add the "set id" bit, which means that the SGID property will be applied.
8. **e.** The script has an error and will not produce the expected output. In a for statement, the loop variable does not have the dollar sign. Changing line 2 to for v1 in a1 a2 will correct the error and produce the output in answer B.
9. **c.** tar should be used with the extraction option x and a tape device, such as SCSI tape /dev/st0.
10. **a.** cp should be aliased to the interactive mode with the -i option in .bashrc. .bash_profile normally doesn't include aliases.
11. **d.** The shadow password system has been implemented, placing all passwords in /etc/shadow as denoted by the x following the username.
12. **PATH.**
13. **d.** The Linux System Administrators' Guide is a free publication of the Linux Documentation Project (LDP) and is available online at <http://www.tldp.org>.
14. **b.** /etc/syslog.conf is the configuration file for the syslog daemon syslogd.
15. **e.** With the top bit of the last byte set in the subnet mask (.128), there are 7 bits left. 27 is 128, less the network address and broadcast address, leaving 126 addresses for hosts.
16. **c.** FTP clients use mget with wildcards.
17. **d.** Routes to the interface and the network are required to exchange information on the local LAN. To act as an Internet workstation (i.e., using Netscape), a default gateway is also necessary.
18. **c.** As defined in /etc/services, port 25 is the SMTP port, often monitored by sendmail.
19. **b.** IP is the underlying datagram protocol.
20. **b.** The ifconfig command is used to configure and display interface information. ipconfig is a Windows utility.
21. **d.** PPP interfaces are not persistent. Instead, they are initialized when needed.
22. **c.** The DHCP client is called pump, after the lady's shoe of the same name. That's an extension from a boot because DHCP is a descendant from bootp.

23. **b.** The spool directory directive looks like this:
sd=/var/spool/lpd/lp
24. **c.** The hosts.deny and hosts.allow files contain configuration information for TCP wrappers. The files won't be used, however, unless tcpd is included in inetd.conf.
25. **c.** Answer a attempts to mount the /proc filesystem. Answers b, d, and e have incorrect syntax.
26. **c.** Both Telnet and FTP are connection-oriented and use TCP for reliable connections.
27. **a, b, and d.**
28. **b.** Read/write access is available to everyone, including root.
29. **d.** NFS-mounted directories seamlessly blend into the local filesystem, requiring no special syntax for access.
30. **c.** While both ICMP and UDP are connectionless, they are different protocols.
31. **c.** CHAP is one of the PPP authentication techniques that embeds its information inside the PPP stream. It is an alternative to clear text passwords.
32. The DNS daemon is **named**. It is included in a package called BIND.
33. **d.** The intent of the chat script is to prepare the modem with appropriate settings to establish a PPP connection.
34. **b.** A print server translates formats, such as PostScript to PCL.
35. **d.** The presence of the localhost address 127.0.0.1 indicates that named is running. Since the system is a workstation, it's safe to assume that it is not serving DNS to a wider community.
36. **a.** BIND v8 has a newer, more modular format, but the information is about the same.
37. **/etc/sendmail.cf.**
38. **/etc/hosts.**
39. **e.** options lines in /etc/modules.conf or /etc/conf.modules configure kernel modules.
40. **e.** lpc is the line printer control program.
41. **c.** Answer a is wrong because it installs modules before compiling them. Answers b and e are wrong because they build the kernel after configuring. Answer d is backward.
42. **c.** tracert is a Windows utility with the same function as traceroute.
43. **b.**
44. **a.** Just as with any interface, routes must be added before communications can occur.
45. **b.** Apache uses httpd.conf, srm.conf, and access.conf. Some implementations may roll all of these files into a single httpd.conf.
46. **b.** The smbprint filter is provided by the Samba package for printing to Windows printers.
47. **c.**
48. **c.** t is the option to list the contents of an archive.
49. The file is /etc/profile.
50. **b.** The .forward file is placed in the home directory containing a single line with the target email address.

51. **d.** Zero exit values usually indicate success.
52. **b.** Instead of using set, the command should have been:
`# export myvar='World'`

This gives the myvar variable to the new shell.

53. **d.**
54. **a.** For security purposes, fingerd is usually disabled using a comment in /etc/inetd.conf.